

Analysts' Earnings Forecast AUC

Following Chen, Cho, Dou & Lev (2022, JAR), Section 4.2.3

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Executive Summary

I compute the AUC of analysts' earnings forecasts using the March restarted GAAP replication sample merged to IBES. Consistent with the instruction, I do not de-drift earnings changes, and I use IBES Actuals street/pro-forma EPS rather than Compustat earnings to measure actual earnings.

The final main sample contains **29,465 firm-years** from **4,503 firms** over fiscal years **2005–2023**. The analyst forecast score has an AUC of **0.8079**, with a 10,000-repetition firm-clustered bootstrap 95% confidence interval of **[0.8024, 0.8135]**. Because this is higher than the Chen et al. (2022) no-drift analyst benchmark of 0.6471, I report additional validation checks. These diagnostics do not find evidence that the result is driven by look-ahead bias, target-year misalignment, AUC implementation, forecast aggregation, IBES currency-field handling, link-score thresholds, EPS rounding, a single fiscal year, or a single industry.

The AUC is higher than the Chen et al. benchmark, but this is not a direct replication of their XBRL sample. I treat the difference as a sample-composition result rather than a replication discrepancy. Coverage composition appears to explain part of the gap, while the remaining difference likely reflects broader sample differences.

1. Design Choices

- **Base sample:** March restarted GAAP replication sample merged to IBES via the WRDS IBES–CRSP historical link table. This is not the paper's XBRL Table 1 sample.
- **Earnings direction:** raw earnings-change direction, with no de-drifting.
- **Actual EPS:** IBES Actuals street/pro-forma EPS (`ibes.act_epsus`, `measure = EPS`, `pdicity = A`).
- **Forecast EPS:** IBES Detail Forecasts (`ibes.det_epsus`, `fpi = 1`, one-year-ahead, `usfirm = 1`).
- **EPS basis:** Actuals and forecasts use the same adjusted IBES EPS basis.
- **Compustat earnings:** no Compustat earnings numbers are used in any AUC direction variable. Compustat is used only to define the replication sample.

2. Sample Construction

Step	Firm-years	Firms
March GAAP replication base sample	50,402	6,396
+ CCM Scheme B permno at formation	45,968	5,908
+ WRDS IBES–CRSP link (score ≤ 2)	44,347	5,763
+ IBES Actual EPS_t known by formation date	40,542	5,499
+ IBES Actual EPS_{t+1} available	39,632	5,431
+ Analyst forecasts in April window	29,465	4,503

Fiscal years 2005–2023; December fiscal-year-end firms only.

AUC-included firm-years have **5.5 times higher median total assets** than excluded firm-years. This is expected because IBES analyst forecasts are concentrated among larger, more visible firms.

3. Methodology

Analyst forecast score. For the main specification, each analyst contributes at most one forecast: the latest forecast issued in the April window. The score is

$$\text{score}_{it} = \frac{\#\{\text{latest analyst forecasts with } \widehat{\text{EPS}}_{t+1} > \text{EPS}_t^{\text{actual}}\}}{\#\{\text{analysts with forecasts in the April window}\}}.$$

Realized direction.

$$y_{it} = \mathbf{1}[\text{EPS}_{t+1}^{\text{actual}} > \text{EPS}_t^{\text{actual}}].$$

Timing. For fiscal year t , the portfolio formation date is March 31 of calendar year $t + 1$. The forecast window is April 1–30 of calendar year $t + 1$, and forecasts target FY $t + 1$ earnings. The all-forecast, paper-literal version produces an AUC of 0.8075, nearly identical to the main per-analyst-latest AUC of 0.8079.

AUC. AUC is computed as the Mann–Whitney rank-sum statistic. Confidence intervals use a 10,000-repetition firm-clustered bootstrap.

4. Main Result

	Value
Main AUC	0.8079
95% CI (firm-clustered bootstrap)	[0.8024, 0.8135]
One-sided bootstrap p -value vs. 0.5	< 0.0001
Firm-years	29,465
Firms	4,503
Fiscal years	2005–2023
Actual increase rate	60.9%
Mean analyst score	0.654
Median analyst score	1.000
Chen et al. (2022) no-drift analyst benchmark	0.6471

The Chen et al. benchmark is reported as a reference point, not as a direct replication target, because this analysis uses the March GAAP replication sample rather than the paper’s XBRL Table 1 sample.

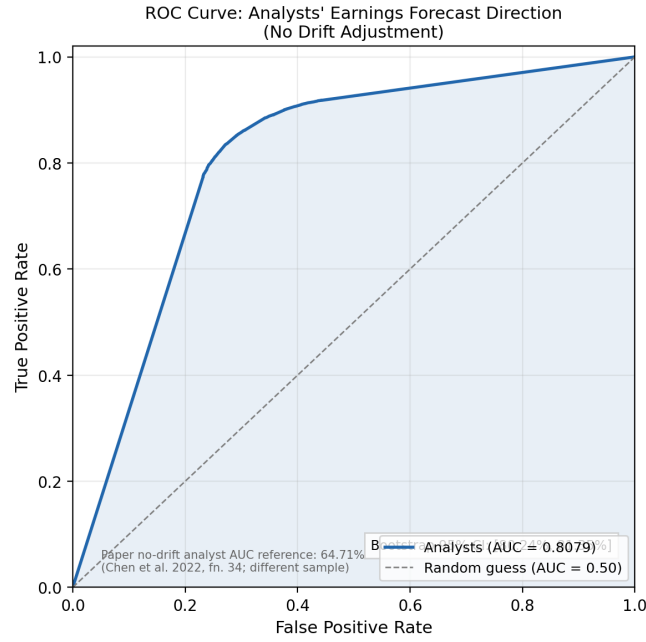


Figure 1: ROC curve for the main analyst forecast score (AUC = 0.8079).

AUC by analyst coverage bucket:

Coverage	N	Share	Increase rate	AUC
1 analyst	6,618	22.5%	57.5%	0.730
2 analysts	4,182	14.2%	58.0%	0.780
3–4 analysts	5,576	18.9%	60.8%	0.811
5+ analysts	13,089	44.4%	63.6%	0.868
All	29,465	100%	60.9%	0.808

Annual AUC ranges from 0.731 (FY2019) to 0.838 (FY2008); it is above 0.5 in all 19 fiscal years. The pooled AUC (0.8079) is close to the simple average of the 19 annual AUCs (≈ 0.796), indicating the result is not a cross-year aggregation artifact. The by-year table is included in the Excel workbook.

5. Pipeline QA

`07_quality_checks.csv`: **14 PASS / 0 FAIL**. These checks cover unique firm-year keys, binary outcomes, score bounds, non-missing IBES Actual EPS, forecast-window timing, forecast fpedats alignment, WRDS link route and score, EPS_t announcement timing, absence of Compustat earnings columns in the final AUC input, IBES EPS basis consistency, and AUC sanity.

6. Additional Validation

Because the AUC of 0.8079 exceeds the Chen et al. benchmark of 0.6471, I ran two additional validation layers. Step 11 forensic validation produced **19 PASS / 0 blocking failures**; the flagged diagnostics are addressed by Step 12. Step 12 pre-submission diagnostics produced **16 OK / 0 WARN / 0 SKIP**.

6.1 No implementation artifacts found

- **AUC formula:** sklearn, Mann–Whitney, and decomposition implementations agree to machine precision.
- **No look-ahead on EPS_t :** 0 violations; EPS_t is announced by the formation date.
- **No look-ahead on EPS_{t+1} :** 0 violations; all forecast dates precede the IBES Actual EPS_{t+1} announcement date.
- **Fiscal target alignment:** 0 mismatches; forecast fpedats align with FY $t + 1$, and the median forecast horizon is 252 days.
- **Aggregation:** per-analyst-latest AUC = 0.8079; all-forecasts version = 0.8075; difference = 0.0004.
- **IBES currency-field handling:** strict `curr_act = USD` versus `USD OR NULL` changes AUC by 0.0000.
- **Link score threshold:** $\text{score} \leq 2$, ≤ 5 , and ≤ 6 produce identical final firm-year sets after IBES Actuals and analyst forecast availability filters.
- **EPS-change tolerance:** applying the same threshold to both realized and forecast directions yields AUCs in $[0.8075, 0.8084]$ for tolerances from 0 to 0.02.

6.2 Robustness checks

Sensitivity	AUC	Δ vs. main
Main	0.8079	—
Positive EPS_t only (N = 24,954)	0.8158	+0.008
Leave-one-year-out range	[0.8029, 0.8100]	$< \pm 0.005$
Leave-one-industry-out range	[0.8048, 0.8109]	$< \pm 0.003$
Excluding 5+ analysts (N = 16,376)	0.7681	−0.040
Only 1 analyst (N = 6,618)	0.7298	−0.078
Equal-bucket observation-reweighted AUC	0.7927	−0.015
Apr 1–7 forecasts only (common sample, N = 12,291)	0.7771	−0.031

Permutation null. The global unconditional permutation null has mean 0.4999 over 3,000 repetitions. The within-year permutation null has mean 0.5302 because pooled AUC combines cross-year pairs while score distributions and realized increase rates differ across years. In both null designs, the observed AUC is far above the null distribution; the upper-bound permutation p -value is < 0.0002 .

7. Score Distribution and Conditional Increase Rates

Score group	N	Share	Actual increase rate
score = 0	7,864	26.7%	18.6%
$0 < \text{score} < 0.5$	1,781	6.0%	33.7%
score = 0.5	834	2.8%	49.9%
$0.5 < \text{score} < 1.0$	2,458	8.3%	65.1%
score = 1.0	16,528	56.1%	83.9%
Not unanimously bullish (score < 1.0)	12,937	43.9%	31.5%

Score bins increase monotonically in actual increase rate.

The immediate source of the high AUC is this strong score separation. When all available analysts’ latest forecasts predict an increase (score = 1.0), the realized increase rate is 83.9%. When the firm-year is not unanimously bullish (score < 1.0), the realized increase rate is 31.5%. This pattern is not only a single-analyst artifact: the score = 1.0 share is 63.9% for 1-analyst firm-years and 52.7% for 5+ analyst firm-years.

8. 2015–2018 Paper-Period Diagnostic

Coverage	N	Share	Increase rate	AUC
1 analyst	1,483	23.5%	59.9%	0.716
2 analysts	894	14.2%	61.6%	0.764
3–4 analysts	1,168	18.5%	67.8%	0.804
5+ analysts	2,773	43.9%	69.7%	0.861
All (2015–2018)	6,318	100%	65.9%	0.794
Chen et al. (2022) no-drift benchmark	—	—	—	0.6471

	AUC
2015–2018, this sample (pooled)	0.7936
2015–2018, 5+ analyst share reweighted to 25%	0.7749
Chen et al. (2022) no-drift benchmark	0.6471

The reweighted AUC uses observation-level weights, giving a proper pairwise-weighted AUC rather than a simple average of coverage-bucket AUCs. Manual cross-check difference = 1.67×10^{-15} .

The 2015–2018 AUC in this sample is higher than the paper benchmark, so the difference is not explained only by using a longer 2005–2023 period. Coverage composition appears to contribute: reducing the 5+ analyst share from 43.9% to 25% lowers the reweighted AUC from 0.7936 to 0.7749. The remaining difference likely reflects broader sample-composition differences between the March GAAP replication sample and the paper’s XBRL 10-K/10-K/A matched sample ($N = 8,149$ submissions).